Usually shoot system of a plant is positively phototropic and negatively geotropic and root system is negatively phototropic and positively geotropic.



Figure 19.2 Negative phototropism (Positively geotropic)

DO YOU KNOW?

Some halophytes produce negatively geotropic roots (e.g. Rhizophora). These roots turn 180° upright for respiration.

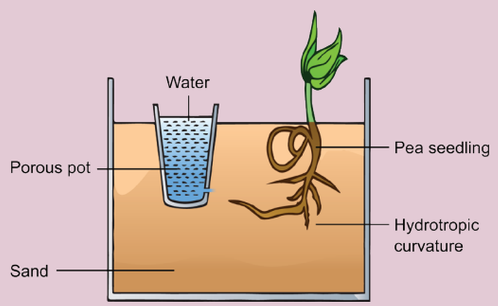


19.2 Nastic Movements

Nastic movements are non-directional response of a plant or part of a plant to stimulus. Based on the nature of stimuli, nastic movements are classified as photonasty and thigmonasty.

Activity 1

Take a glass trough and fill it with sand. Keep a flower pot containing water, plugged at the bottom at the centre of the glass trough. Place some soaked pea or bean seeds around the pot in the sand. What do you observe after 6 or 7 days? Record your observation.



Activity 2

Take pea seeds soaked in water overnight. Wait for the pea seeds to germinate. Once the seedling has grown put it in a box with an opening for light on one side. After few hours, you can clearly see how the stem has bent and grown towards the light.

Photonasty: Movement of a part of a plant in response to light. e.g.*Taraxacum officinale*, blooms in morning and closes in the evening. Similarly, *Ipomea alba* (Moon flower), opens in the night and closes during the day.



Night



Day

Figure 19.3 Photonasty in Moon flower

Thigmonasty: Movement of a part of plant in response to touch. e.g. *Mimosa pudica*, folds leaves and droops when touched. It is also known as Seismonasty.



Figure 19.4 Thigmonasty in Mimosa pudica

DO YOU KNOW?

The Venus Flytrap (*Dionaea muscipula*) presents a spectacular example of thigmonasty. It exhibits one of the fastest known nastic movement.





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